

First Handout: PHY 461&462 Experimental Physics (2023-24 sem-II)

This is a lab-based course. The laboratory is equipped with experiments in condensed matter physics, optics, nuclear physics and spectroscopy. Students will perform experiments assigned to them twice a week in M.Sc. lab (NCL 206 & 207).

Laboratory Days & Duration:

Tuesday and Wednesday (2 PM – 6 PM) ----Group A

Thursday and Friday (2 PM – 6 PM) ---Group B

Reading material: Laboratory Manual and references therein. Lab Manual will be available at Copy Point.

Grading of Laboratory Reports:

Students must submit the lab report within one week of completing the experiment. It should contain a brief introduction to the theory of experiment, principle involved (ray diagram/ circuit diagram/ sketch etc., methodology of performing the experiment, data table, data analysis, results, error analysis.

Grading scheme (Lab reports must be graded within one week of submission.)

Grading schemes for lab report of each experiment:

Total: 40 Marks

1. Theory + Formula + Diagram(circuit/ray/block): 10
2. Data: 10
3. Result with error analysis: 15 (10+5)
4. Viva = 5

Daily Evaluation (Punctuality, Preparedness, discipline and performance in the lab): 10 Marks.

End Sem Evaluation: There will be end sem examination (Experiment + Viva).

Final Grades will be based on the following two components:

- (i) Performance throughout the semester-50%
- (ii) Final Lab Evaluation-50%

Each student will perform 8 experiments.

Course Content:

1. Error Analysis
2. Stefan's Constant
3. Millikan's Oil Drop Experiment
4. Bragg's Diffraction with Microwaves
5. Magnetic Hysteresis
6. Faraday Rotation
7. Electron Spin Resonance

8. Michelson's Interferometer using Sodium Lamp
9. Michelson's Interferometer using He-Ne Laser
10. Diffraction of He-Ne Laser Beam by 2D Objects
11. Concave Grating Spectrometer, Rowland Mounting
12. Iodine Absorption Spectrum
13. Zeeman Effect
14. Fluorescence of Uranyl Nitrate Hexahydrate (U.N.H)
15. Hall Effect
16. Magnetoresistance and its Field Dependence
17. Electrical Resistance of Superconductor
18. Ionic Conductivity of Alkali Halide
19. G. M. Counter, Counting Statistics and γ Absorption Cross Section
20. Proportional Counter
21. Gamma Ray Spectrometer
22. X-Ray Diffraction
23. Nuclear Magnetic Resonance
24. Study of Thermoluminescence of Colour Centers in Crystals
25. Electrical Conductivity and Bandgap Measurements
26. Photoconductivity Experiment
27. Characteristics of Microwave Generator (Klystron)
28. Study of Gunn Oscillator
29. Electro-Optic Effect
30. Phase Transition in BaTiO_3
31. AC Susceptibility of a High T_C Superconductor

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